

25/12/2025 Christmas

classmate

Date

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DevOps & Cloud

* Devops Roadmap 2026 *

* Phases of Learning *

Phase 1 (Foundation)	Phase 2 (Shipping)
Linux, Bash, Scripting Networking, Git & Team Workflow	Containers, Cloud Fundamentals, CI/CD, ^{Azure DevOps} Docker, Github Action, Jenkins
Phase 3 (Scaling)	Phase 4 (Production)
IaC, Kubernetes, Helm, Terraform, Ansible ArgoCD, GitOps	Observability, DevSecOps, Incident Response, Agentic AI, Grafana, Prometheus, ^{System Design}

Phase 5 (MLOps) (Intelligence)

Data Versioning with DVC, Experiment Tracking
MLFlow, Kubernetes, Kserve
AWS Sagemaker AI, Vertex AI
CI/CD, Kubeflow, Docker,
FastAPI, Model Deployment, Model
Serving, Agentic AI, RAG,
Monitoring, Data Drift, Feature Engi-
neering, Feature Store, Model Training

PHASE 1 (Foundation)

1) Linux → Thinking like System Engineer

- File System hierarchy (`/`, `/var`, `/etc`, `/proc`)
- File types, inodes, symbolic v/s hard links
- Users, groups, permissions, `umask`
- `sudo` internals & security implications
- Process lifecycle (`fork`, `exec`, signals)
- `systemd`:
 - units, targets, dependencies
 - service failure & restart loops
- Logs:
 - `journalctl` deep usage
 - apps logs vs System logs
- Disk:
 - mount points, `fstab` awareness
 - disk pressure & inode exhaustion
- "Memory" & CPU:
 - free, `vmstat` basics
 - CPU load v/s CPU usage
- SSH
 - key-based auth
 - port forwarding use cases
- Common failures:
 - service not starting
 - permission denied
 - disk full
 - high load

2) Bash & Automation Mindset / Python

- Shell execution Model
- Exit codes & error handling
- Variables, quoting, conditionals
- Loops & Functions
- Positional parameters
- Pipes and redirection
- Cron & Scheduling pitfalls
- Idempotent scripting mindset
- Logging in scripts
- Defensive scripting

3) Networking (Absolute Basics)

- OSI vs TCP/IP (Practical mapping)
- IP addressing & CIDR
- Ports & sockets
- TCP handshake & common failures
- DNS : ~~→~~ request lifecycle → headers, status codes
- HTTP : → resolution flow → caching & TTL
- HTTPS : → certificates → TLS termination
- Load Balancing : → L4 vs L7 → Health checks
- Debugging tools: curl, dig/nslookup, ss/netstat, traceroute, tcpdump (basic)

4) Git & Team Workflows

- Git internals, Branching strategy (Green is feature)
- Merge vs Rebase (when & why)
- Conflict resolution, GitHub / GitLab / BitBucket
- PR review, etiquette, Tagging & releases
- Commit hygiene, README writing (Engineering communication)

PHASE 2 (Shipping Software)

1) Containers (Docker Deep Reality)

- Why containers exist (VM Pain)
- Image vs Container vs Runtime
- Dockerfile
 - layer caching
 - ENTRYPOINT vs CMD
 - env handling
- Multi Stage Builds
- Image size & Attack surface
- Volumes and Bind mounts
- Docker Networking
- Docker Compose Patterns (Scenario)
- Common failures:
 - container exits
 - port not exposed
 - env misconfig
 - dependency missing

2) Cloud Fundamentals (AWS/Azure/GCP)

- Shared responsibility model
- IAM: → user vs roles → policies → access boundary
- EC2: → Instance lifecycle → security groups
- VPC: → CIDR Planning → Public vs Private subnets
 - Routing
- Storage: → S3 basics and Policies
- Databases: → RDS Awareness
- Registry: → ECR Workflow
- Cost Awareness: → Common mistakes.

3) CI/CD (Delivery Engineering)

- CI/CD (CI vs CD)
- Pipeline Design Principles
- Pipeline as Code
- Build → test → scan → artifacts
- Secrets Management
- Deployment strategies:
 - Rolling
 - Blue-Green
 - Canary (conceptual)
- Rollback Mechanisms
- Failure Handling
- Pipeline Observability (Logs/Artifacts)
- Tools → Jenkins
 - Github Action
 - GitLab

4) Azure DevOps (Enterprise Awareness)

- Why Enterprises use it
- Pipeline YAML Overview
- Repo + Boards Basic
- Comparison with Github actions

PHASE 3 (Cloud At Scale)

1) Infrastructure as Code & Configuration Management (Terraform & Ansible)

- Declarative vs Imperative
- Providers & Resources
- Variables & outputs
- State: → why it exists → drift
- Modules: → structure → Reusability
- Environment Separation (Workspaces)
- Best practices: → naming → versioning
- Remote Backends:
- Providers & Modules
- Ansible Playbooks
- Ansible Templates | Roles | Galaxy

2) Kubernetes (Operational, Not Theoretical)

- Why Kubernetes exists (CKA)
- Latest versions | Migrations | Upgrade (v1.35)
- Cluster architecture (high-level)
- Pods & Controllers
- Deployments vs StatefulSets
- Services & Networking
- Ingress
- ConfigMaps & Secrets
- Storage: → PVCs
- Reliability: → probes → resource limits
- kubectl: → describe → logs → exec
- Common failures: → CrashLoopBackOff, image pull errors, misconfigured services

3) Helm & GitOps

- Helm chart structure
- Values & templating
- Helm upgrades & rollbacks
- GitOps principles
- Argo CD:
 - sync
 - drift correction
 - rollback via Git
- End to end CI/CD Pipelines

PHASE 4 (Production Readiness)

1) Observability

- Metrics vs logs vs traces
- RED & USE Signals
- Alerting philosophy
- Prometheus basics
- Grafana dashboards
- Logging patterns
- Open Telemetry: → Why it exists → traces flow
- Debugging with data

2) DevSecOps (Practical Security)

- Secrets hygiene
- Image Scanning
- Dependency awareness
- K8s RBAC basics
- Least privilege mindset
- Supply chain awareness (SBOM)

3) Incident Response & SRE Basics

- On-call responsibility
- Incident lifecycle
- Postmortems
- Runbooks
- Reliability mindset.

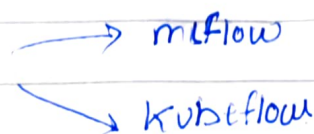
4) Job Readiness

- Resume with engineering impact.
- Github as Portfolio.
- Interview story Building.
- Debugging interviews
- System design basics (entry level)

5) Agentic AI for DevOps

- Building Agents with Python.
- Deploying Agents at Scale.
- MCP Integration.
- Tools & Gateways
- Project: DevOps Copilot Agent that reads logs or CI failures and suggests fixes.
- AIOps on AWS

6) MLOps



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graph LR; MLOps --> mlflow; MLOps --> kubeflow
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